

Eugene Lafayette Buchanan

Senior Software Engineer and Technical Consultant

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Professional Summary

Senior software engineer with 25+ years building, shipping, implementing and supporting solutions, from streaming media and operating systems to learning management integrations and agentic AI infrastructure. Expert in quality assurance (QA), test automation, and integrations for ERP, education, and streaming platforms. I also bring hands-on IT operations: campus support, retail service desk, ticket queues, and software evaluation and migration. I write clean code, build reliable test systems, and explain complex ideas clearly.

Skills

Languages and Frameworks: JavaScript, TypeScript, Python, Rust, PHP, Node.js, React, SQL

Testing and QA: Cypress, Selenium, Postman, JUnit, Qase, white-box testing, regression suites, performance and load testing, accessibility testing, security testing, usability and localization testing

Cloud, CI/CD and DevOps: AWS (ECS, S3), Apache, CI/CD pipelines, Docker, Git, Bitbucket, Elasticsearch, Unix domain sockets, JSON-RPC

Integrations and Auth: SAP, Oracle, Workday, Coursera, UKG, Zoom, SAML, OAuth, SFTP, REST APIs, webhooks, SCORM, LTI

Collaboration and Tooling: Slack, Jira, Confluence, Scorm Cloud

AI and Automation: AI automation, AI support chatbot with retrieval-augmented generation (RAG) and guardrails, prompt engineering, large language model (LLM) integration, agentic AI, LangGraph, Model Context Protocol (MCP)

Support and Operations: ticket triage and queue support, retail service desk, campus IT support, security and backup policies, wireless and video surveillance, software evaluation and migration, IT asset tracking

Compliance and Quality Practice: SOC 2 compliance research and implementation, Blue Team incident forensics, technical debt reduction, Standard Operating Procedure (SOP) guides, technical writing

Platforms and Domains: ERP, CRM, learning management systems (LMS), streaming media, enterprise software, alternative energy, real estate

Leadership and Communication: Team building, mentoring, training, cross-functional collaboration, product ownership, backlog management, PDR life cycles, presales engineering

Work Experience — Software Engineering

Senior QA Engineer / Test Automation Lead / Product Owner — Integrations

Knowledgecity LLC, Remote, Dec 2020 – Aug 2025

- Led QA across integration, frontend, backend, mobile, security, performance, load, API, database, functionality, usability, accessibility, and localization testing. Reduced technical debt and joined Blue Team incident forensics.
- Built and maintained a CI/CD quality pipeline (Apache, REST, SQL, AWS, React) with Cypress, Selenium, Postman, JUnit, Elasticsearch, and S3. Used Slack, Jira, Bitbucket, Confluence, Git, Node, JavaScript, PHP, Scorm Cloud, Elastic, and Qase.
- Hired and trained the test automation team. Built end-to-end regression suites with health dashboards and alert systems for complex deployment scenarios.
- Shipped integrations with SAP, Oracle, Workday, Coursera, UKG, and Zoom using SAML, OAuth, SFTP, custom APIs, webhooks, SCORM, and LTI.
- Managed dev and test teams, the product backlog, and PDR life cycles with KPI tracking as Integrations Product Owner.
- Helped develop an AI support chatbot with retrieval-augmented generation (RAG) context retrieval and guardrails. Researched and implemented SOC 2 compliance.

Business Consultant — Campus IT and Community Programs

Medico Investments LLC / Foremost Senior Care / Foremost Organization, Hesperia, CA, 2013 – 2016

- Provided campus IT support: wrote security and backup policies for campus servers and PCs, upgraded video surveillance for better quality and remote access, and installed campus-wide wireless. Set up sound and video equipment and MC'ed events.
- Evaluated and migrated accounting and medical-records software.
- Built the nonprofit and community outreach program: set up the bingo license, filed federal, state, and city applications, and ran a community social program with dance lessons.
- Recovered a non-permitted photovoltaic installation and worked with the city for permission to operate using as-built engineering.
- Reviewed business plans and cut procurement, operations, and staffing costs. Won city approval for a bridge to landlocked land.

Computer Service Technician — Geek Squad

Best Buy, Victorville, CA, 2007

- Ran the retail service desk: logged tickets, triaged walk-in customers, tracked units through check-in, repair, and pickup, and gave plain-language status updates.
- Set up new PCs, migrated user data, and fixed virus, operating system, and hardware issues, while recommending service plans, accessories, and upgrades. Handled upselling, customer service, payments, and refunds.

Freelance IT Consultant

Amp Computer, Apple Valley, CA, 2003 – 2011

- Sold and supported servers and PCs. Configured ERP and CRM systems, office automation, and custom solutions.

IT Manager

Lucerne Valley Unified School District, Lucerne Valley, CA, 2002

- Provided district-wide support across four school campuses, including WAN/LAN, NetWare servers, Windows, Linux, and virtualized desktops, plus audio/video equipment.
- Met state attendance submission requirements, running data extract, transform, and load (ETL) with database table locking and certification.

Senior Software Engineer II

RealNetworks, Seattle, WA, 1999 – 2001

- Ran Software Development Engineer in Test (SDET) quality assurance across Linux, Unix, Windows, and embedded systems for a streaming media platform.
- Served on an advanced research team focused on consumer appliances, mobile platforms, stream servers, and cellular networks, including TFRCP patent work.
- Provided global presales engineering for consumer devices with vice president sales support, including travel to Japan, Korea, and the continental United States. Trained new hires.

Earlier Software Test Engineering and Support Roles

Microsoft, IBM, Keene Inc., 1996 – 1999

- Microsoft, Redmond, WA (1999): Senior Software Test Engineer IV, Windows Media Server versions 4 and 5; nightly build test script automation.
- IBM, Kirkland, WA (1998): Senior Test Engineer III for high-availability servers; WHQL certification for cluster failover.
- Microsoft, Redmond, WA (1997): Software Test Engineer II for Windows 98 and Windows NT; original equipment manufacturer (OEM) setup and hardware/driver verification.
- Keene Inc., Seattle, WA (1996): Technical Support Agent II for Windows 95 and Windows NT; frontline and escalated support ticketing.

Work Experience — Additional Professional Roles

Alternative Energy Consultant

2004 – Present

- PhaseSave.com, Palm Springs, CA (2017 – 2021 / Present): Partner and Director of research and development, sales engineering, project management, product and vendor sourcing, and contractor management. Expertise in solar power, air conditioning, phase-change materials, cold chain logistics, finance, and tax credits.
- EWSolar.net LLC, Homeland, CA (2014 – 2016): Vice President, pioneering large-scale solar-powered water pumping projects for commercial and municipal customers.
- Free Energy Resources Inc., Phelan, CA (2013): Chief Technology Officer of a Korean solar marketing company; sales, procurement, project management, permitting, and finance.
- GRIDNOT Inc., Lucerne Valley, CA (2007 – 2012): Vice President, sales and integration of solar, wind, and geothermal air-conditioning systems.
- OnPoint Power Inc., Norco, CA (2006): Partner, residential and light-commercial solar sales; pioneered a horse-manure waste-to-energy plant proposal presented to city council meetings.
- Desert Power Inc., Palm Springs, CA (2005): Executive Assistant, sales and engineering support for solar, waste-to-energy, and tri-generation projects; research on Fischer-Tropsch syngas conversion, process stabilization, and absorption refrigeration.
- Solatron Inc., Victorville, CA (2004): Sales Representative, inbound phone sales of photovoltaic systems; built an Excel calculator adopted office-wide and worked on SOP sales scripts.

Founder — Land Technology

Tierrachain.com, 2023 – Present

- Founder of a Mexican land corporation building software as a service (SaaS) that uses AI and blockchain to improve real estate exchange in coastal exclusion zones.
- Developing a parallel public records system with AI-automated compliance filing for arbitrage, plus fintech extensions using NFTitle and Bitso payment rails on a real estate mapping site.

Elected Board President

Mariana Ranchos County Water Board, 2006 – 2010

- California public officer and elected board president, the youngest in California history; versed in public ethics, Roberts Rules of Order, the Brown Act, and state and federal agency law. California JPIA delegate with public accounting, insurance, and safety expertise.
- Cut budget waste by 53%, replaced corrupt personnel, hired a strong general manager, and blocked a high-risk floodplain stack-and-pack project that later flooded.

Open-Source Contributor — Agentic AI Infrastructure

ZeroClaw Labs, github.com/zeroclaw-labs/zeroclaw, 32.8k GitHub stars, 2026 – Present

- PR #7946 (merged Jul 2026): added the model context window meter bar across the zerocode TUI, gateway agent chat, and command-line interactive mode, with one `context_window` source of truth covering 9 providers.
- PR #8966 (open): carrying live provider identity on usage events so usage, cost, and context-meter attribution follow the serving provider across fallback and model switches.
- PR #8337 (closed, superseded by #10269): built an observer reporting agent lifecycle states over JSON-RPC on Unix domain sockets; cleared four rounds of maintainer review, and its requirements were adopted into the vendor-neutral lifecycle design.

Education

- Associate Degree, Victor Valley College
- California Notary Commission
- Toastmasters International
- High School Diploma, Lucerne Valley High School